

DR-AGENTOS: PREDICTIVE 4D BELIEF FOR NAVIGATION IN EVOLVING WORLDS

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ABSTRACT

Embodied navigation is commonly formulated over a stationary world snapshot: maps and memories are assumed to describe the same state in which actions are executed. This abstraction breaks for long-lived agents. During periods without observation, objects are relocated, spaces are reconfigured, and accessibility changes, so historically valid evidence may induce obsolete goals and inefficient routes. We study *persistent navigation in evolving worlds*, where an agent must infer the current task-relevant world state from partial and temporally stale observations, act under the resulting uncertainty, and revise its plan as physical evidence arrives. We introduce DR-AGENTOS, a predictive embodied-agent architecture whose navigation core, P4D-NAV, transforms an evidence-grounded 4D memory into a belief over the current world state. A continuous-time Transformer encodes irregular histories, while a structured persistence–relocation pointer head estimates whether the last supported state remains valid and, otherwise, distributes probability over plausible alternatives. A visibility-aware Bayesian update converts positive and negative observations into posterior evidence, enabling direct navigation under concentrated belief, cost-aware exploration under uncertainty, and replanning when memory and observation disagree. We further introduce P4D-BENCH, a 54-scene, 90-scene-day benchmark with 803.68k tasks and 239.19k hidden state changes, spanning state prediction, object navigation, vision-and-language navigation, and embodied question answering. Paired static, routine, and random worlds isolate whether gains arise from learned dynamics rather than scene-frequency shortcuts. Together, DR-AGENTOS reframes embodied navigation from executing against a remembered snapshot to reasoning and acting over a continuously evolving world state.

1 INTRODUCTION

Embodied navigation requires an agent to ground a goal, maintain a world model, and select actions from egocentric observations. Most formulations nevertheless adopt a snapshot abstraction: the map or memory is assumed to describe the same world state in which navigation is executed (Chaplot et al., 2020; Huang et al., 2023; Gu et al., 2024). Long-lived agents violate this assumption. Homes, offices, and campuses evolve while unobserved as objects are relocated, spaces are rearranged, and accessibility changes. The agent may therefore retain evidence that is historically correct yet decisionally stale, pointing to an obsolete goal or an inefficient route.

Long-term deployment creates a dual tension. A persistent agent is not a persistent observer: as its *4D history* grows, individual records become stale, yet repeated experience reveals environmental dynamics, routines, and personal habits. More history therefore does not automatically mean better knowledge of the present. Meanwhile, history grows without bound, whereas decision-time context and computation must remain controlled. Most embodied memories are retrospective, organizing or retrieving what was observed (Liu et al., 2024; Kurenkov et al., 2023); a predictive memory should instead infer *what task-relevant state is most likely true now, how uncertain that belief is, and how navigation should adapt*. We seek a decision-preserving interface in which the episodic store retains grounded facts while the action policy consumes a compact, task-conditioned belief that preserves prediction and search utility.

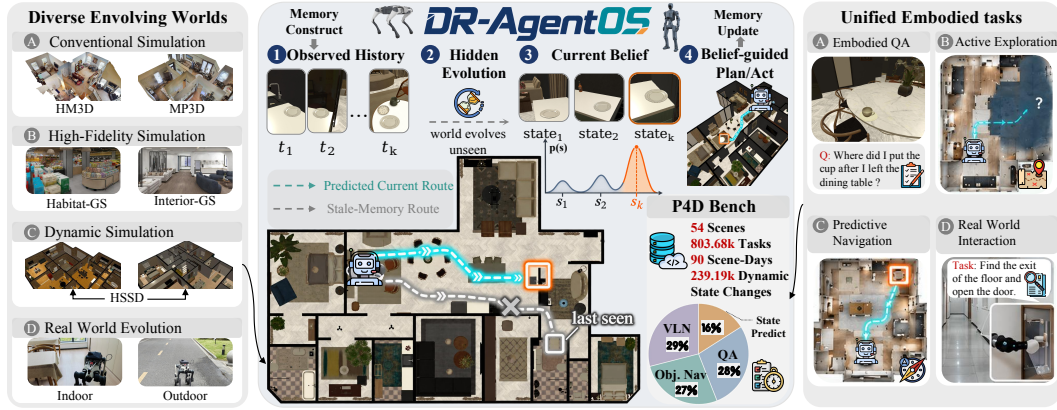


Figure 1: Overview of DR-AGENTOS in evolving worlds. Observed histories are transformed into current-state beliefs that guide planning and are updated by new evidence, supporting prediction, navigation, exploration, question answering, and real-world interaction.

Recent work models temporal persistence, future relations, household routines, and targets that move during navigation (Saavedra-Ruiz et al., 2026; Patel & Chernova, 2023; Argenziano et al., 2026; Dorbala et al., 2026). These approaches establish that dynamics matter, but a complementary gap remains: navigation must reason through periods in which world-state changes are not observed, rather than merely update a map after they become visible. How should an agent transform its own growing, partial history into a predictive belief that directly controls action and is corrected by evidence collected along the route?

As illustrated in Figure 1, DR-AGENTOS places this predictive loop in a unified embodied system. It constructs memory from conventional simulation (HM3D and MP3D), high-fidelity simulation (Habitat-GS and Interior-GS), dynamic HSSD scenes, and indoor or outdoor real-world observations. When the world evolves out of view, the agent transforms observed history into a belief over the current state, uses that belief to plan and act rather than blindly following a stale-memory route, and writes newly collected evidence back to memory. The same interface supports embodied question answering, active exploration, predictive navigation, and real-world interaction. P4D-BENCH provides the controlled simulated evaluation, while the real-world panel illustrates the deployment interface rather than a benchmark result.

We formulate the broader setting as **persistent navigation in evolving worlds**: before a query, an agent has accumulated timestamped observations through intermittent visits, while the task-relevant world state may have changed during the resulting observation gaps. We operationalize its central uncertainty through **persistent dynamic object navigation**, where a portable target may undergo hidden transitions after its last positive observation. The agent receives neither the current target state nor the activity that caused the change and must navigate under a fixed budget using only causal memory, candidate receptacles and viewpoints, and online RGB-D observations. Our main protocol freezes the target after the query to isolate reasoning over stale history; an extension allows another hidden transition during navigation.

We introduce DR-AGENTOS, a long-term embodied agent centered on **4D Predictive Memory**, and instantiate its predictive-navigation core as P4D-NAV. Its 4D store maintains persistent identities, time-valid states, and observation provenance in a common frame. A causal retriever selects target-relevant history, and a lightweight continuous-time Transformer represents irregular gaps and calendar periodicity. A structured head factorizes the current belief into the probability that the last-seen state persists and a pointer distribution over relocation candidates. This matches the navigator’s physical alternatives, supports variable candidate sets, and exposes the belief—rather than the event sequence—as the memory interface to the action policy.

Prediction is embedded in a closed decision loop. Concentrated beliefs support a direct attempt; diffuse beliefs induce probability–cost-aware exploration. Navigation returns new RGB-D evidence, and a visibility-aware Bayesian update suppresses an inspected hypothesis only when the target should have been observable there. The agent then replans from the posterior. The same controller

handles rare, routine, personal, and irregular changes without an oracle dynamics label, connecting history $\rightarrow$ current-state belief $\rightarrow$ plan/action $\rightarrow$ evidence $\rightarrow$ memory update.

We evaluate this capability with P4D-BENCH, a 54-scene benchmark containing 90 scene-days, 239.19k dynamic state changes, and 803.68k task instances across state prediction, object navigation, vision-and-language navigation, and embodied question answering. Its evolving HSSD histories combine longitudinal activity records, real human–object interaction traces, and household state-transition programs. Our controlled analysis measures belief quality, navigation efficiency, evidence-aware recovery, and generalization. Paired static, routine, and random worlds share the observed setup while differing only in the hidden transition process, testing whether gains arise from learnable temporal structure rather than scene frequency or route geometry.

Contributions. Our contributions are threefold:

- We formulate **persistent navigation in evolving worlds**, exposing long-term history as both stale evidence about prior states and a source of regularities for inferring the present; persistent dynamic object navigation provides a controlled operationalization.
- We introduce DR-AGENTOS and its P4D-NAV core with **4D Predictive Memory**, which transform causal history into an action-facing persistence–relocation belief and close the loop through physical evidence, memory update, and replanning.
- We introduce P4D-BENCH, a human-trace-grounded, 54-scene benchmark spanning state prediction, ObjectNav, VLN, and EQA with paired static/routine/random controls.

2 RELATED WORK

Embodied navigation and active search. Embodied navigation spans object-goal, language-conditioned, and exploratory tasks that couple semantic perception, spatial reasoning, and action (Anderson et al., 2018; Savva et al., 2019; Chaplot et al., 2020; Xin et al., 2026). Semantic maps and open-vocabulary scene representations improve generalization by associating language with metric or graph-structured space (Huang et al., 2023; Gu et al., 2024; Werby et al., 2024). Probabilistic object-search methods use object–room or object–object correlations to order exploration and update a belief after observations (Zheng et al., 2022; Ginting et al., 2024). These formulations generally assume that the task-relevant world state remains consistent with the map during an episode. P4D-NAV addresses a different source of uncertainty: historically valid observations can become decisionally stale because the environment evolves between visits.

Navigation in changing environments. Early work on dynamic mapping often filters moving content or updates a semantic map after changes are observed (Bescos et al., 2018; Narayana et al., 2020). Scene Graph Memory models a partially observable dynamic graph and predicts object–location links (Kurenkov et al., 2023), while DynaMem continually reconstructs an actionable spatio-semantic memory for mobile manipulation (Liu et al., 2024). Predictive spatio-temporal scene graphs attach persistence and emergence models to open-vocabulary relations (Saavedra-Ruiz et al., 2026). FlowMaps learns multimodal continuous future object placements and uses ranked hypotheses for dynamic ObjectNav (Argenziano et al., 2026). Portable Object Navigation and Transit-Aware Planning model objects that move as the agent moves (Dorbala et al., 2026). Our setting is complementary: dynamics are inferred from the agent’s causal, intermittent history; object-specific regimes are hidden; and prediction is embedded in an observe–update–replan loop rather than consumed only as a fixed candidate ranking or known transition path.

Long-term embodied memory. Long-horizon benchmarks increasingly require agents to retain observations across exploration or interaction (Majumdar et al., 2024; Yadav et al., 2025; Zhang et al., 2026). Open-vocabulary 3D memories support spatial retrieval (Gu et al., 2024; Werby et al., 2024), while recent 4D representations index time-varying states or human-centered events (Sohn et al., 2025; Bartoli et al., 2026). These systems motivate persistent identity, temporal validity, and provenance. We use these properties as infrastructure, but focus the learning problem on a navigation-relevant belief: whether an entity’s last state persists and, if not, which reachable state should be inspected next.

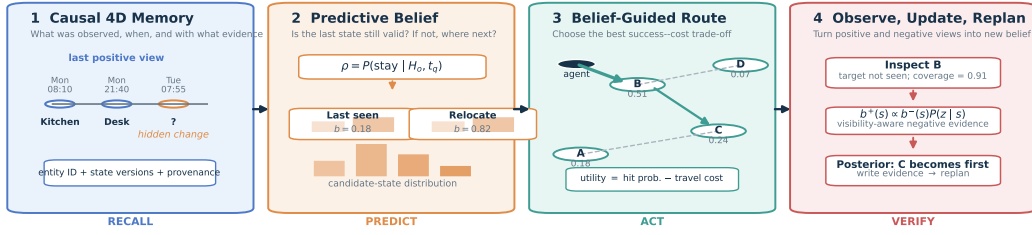


Figure 2: Predictive-navigation core of DR-AGENTOS. A causal 4D memory is propagated to the query time as a structured persistence–relocation belief. The agent trades target probability against travel cost, converts visibility-qualified observations into posterior evidence, and replans after unsuccessful inspections.

Human routines and personalized object dynamics. Household activities exhibit temporal and person-specific regularities. HOMER and SLTe-PRO predict routine object usage for proactive assistance (Patel & Chernova, 2023); STREAK studies continual adaptation under household context drift (Bartoli et al., 2024); and UCON evaluates navigation with a large user-habit knowledge base (Wang et al., 2026). Existing dynamic-world generators commonly rely on hand-designed routines or language-model priors. P4D-BENCH instead normalizes complementary longitudinal activity, real object-interaction, and complete state-transition records into executable HSSD episodes. Its paired worlds preserve marginal scene and query factors while changing the transition mechanism, enabling controlled measurement of when routine learning helps navigation.

3 PREDICTIVE 4D BELIEF NAVIGATION

3.1 PERSISTENT NAVIGATION IN EVOLVING WORLDS

At the system level, the task-relevant state may include object placement, spatial configuration, and accessibility. To isolate the coupling between hidden world evolution, predictive belief, and physical action, we formalize the core task through portable-object goals. Let an environment contain a navigable space $\mathcal{X}$, a set of persistent entities $\mathcal{O}$, and candidate object states $\mathcal{C}_o$ for each target $o \in \mathcal{O}$. A state identifies a semantically valid receptacle or region and its associated inspection viewpoints. Before a query at time t_q , the agent has collected only a partial history

$$\mathcal{H}_{\leq t_q} = \{(I_k, D_k, \xi_k, t_k) \mid t_k \leq t_q\}, \quad (1)$$

where I_k, D_k, ξ_k denote RGB, depth, and camera pose. The target’s true query-time state $s_q^o \in \mathcal{C}_o \cup \{\text{unknown}\}$ and any unobserved transition events are hidden.

An episode supplies a target description, query time, initial agent pose x_0 , a causal memory materialized from $\mathcal{H}_{\leq t_q}$, and a finite action budget. At decision step j , the agent chooses a candidate viewpoint or a low-level navigation action, receives an observation z_j , and may terminate with STOP or NOT_FOUND. Success requires a correct stop near a valid goal viewpoint with the target visible. The policy should maximize success while minimizing travelled distance and inspections:

$$\pi^* = \arg \max_{\pi} \mathbb{E}_{\pi} [\mathbb{I}(\text{success}) - \lambda_d L - \lambda_n N_{\text{inspect}}]. \quad (2)$$

The main protocol holds the target fixed after t_q , isolating whether a stale history can be transformed into a useful current belief. The online-dynamic extension advances the world clock during execution and may apply an additional transition while the target is outside the camera view.

3.2 EVIDENCE-GROUNDED 4D MEMORY

The 4D memory represents a persistent 3D semantic scene whose entity states evolve over time. For image coordinate $\tilde{\mathbf{u}} = [u, v, 1]^\top$ with depth $D_t(u, v)$, the corresponding world point is

$$\mathbf{x}^w = \text{proj}_3 \left(\mathbf{T}_{w \leftarrow c, t} \begin{bmatrix} D_t(u, v) \mathbf{K}^{-1} \tilde{\mathbf{u}} \\ 1 \end{bmatrix} \right), \quad (3)$$

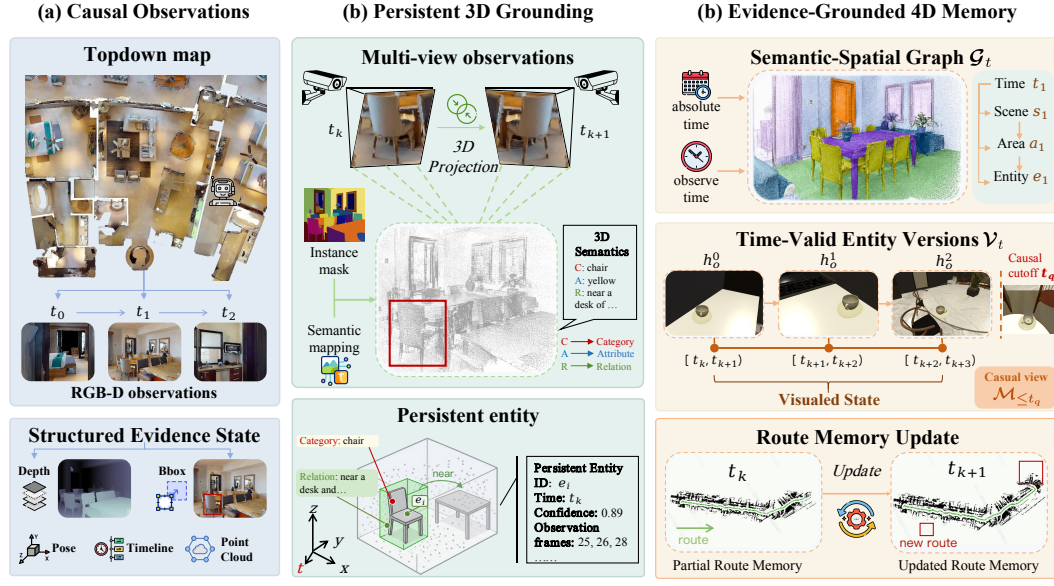


Figure 3: 4D memory construction. Multimodal observations are registered in a shared frame, consolidated into persistent entities, and retained as temporally versioned states with supporting evidence.

where $\mathbf{K}$ is the camera intrinsic matrix and $\mathbf{T}_{w \leftarrow c, t}$ is the camera-to-world transform. Cross-view association consolidates observations using semantic compatibility, 3D overlap, temporal consistency, and source confidence. Repeated observations strengthen one persistent entity rather than creating disconnected detections.

Memory is materialized causally from versioned deltas:

$$\mathcal{M}_{\leq t} = \text{Materialize}(\Delta\mathcal{M}_0, \dots, \Delta\mathcal{M}_t) = (\mathcal{G}_t, \mathcal{V}_t, \mathcal{B}_t), \quad (4)$$

where $\mathcal{G}_t$ stores semantic and spatial relations, $\mathcal{V}_t$ stores time-valid entity versions, and $\mathcal{B}_t$ stores observation provenance. The m -th version of entity o is

$$h_o^m = ([t_{o,s}^m, t_{o,e}^m], s_o^m, c_o^m, f_o^m, \mathcal{P}_o^m), \quad (5)$$

containing a validity interval, candidate state, confidence, visual feature, and evidence handles. Persistent identity is separated from mutable state: moving an object appends a version without erasing its earlier trajectory. All query-time views exclude observations and events after t_q .

3.3 TARGET-CONDITIONED HISTORY RETRIEVAL

For a target o , the retriever first resolves its persistent identity or category-compatible instances, then returns the most recent K causal events

$$H_o = \{e_k = (o_k, s_k, t_k, z_k, c_k, v_k)\}_{k=1}^K. \quad (6)$$

Here s_k is the observed candidate state; $z_k \in \{\text{seen}, \text{notseen}\}$ is the evidence type; c_k is perception confidence; and v_k summarizes whether the hypothesized state was adequately visible, including frustum coverage and unoccluded fraction. Context events may be included when they share an entity, receptacle, region, or time window with the target, but hidden activity labels are never exposed at test time.

Candidates $i \in \mathcal{C}_o$ are not global class indices. Each is encoded from its region, receptacle category, functional role, geometry, and instance identity. This pointer-compatible representation permits the number and identity of states to vary across homes and supports evaluation on unseen receptacle instances. An additional `unknown` candidate absorbs states outside the mapped set.

3.4 CONTINUOUS-TIME HISTORY ENCODER

Each event token combines entity, state, evidence, confidence, and visibility embeddings with an explicit encoding of elapsed and calendar time:

$$\mathbf{x}_k = E_o(o_k) + E_s(s_k) + E_z(z_k) + E_c(c_k) + E_v(v_k) \quad (7)$$

$$+ \phi_\Delta(t_q - t_k) + \phi_{\text{cal}}(t_k), \quad (8)$$

where ϕ_Δ includes $\log(1 + t_q - t_k)$ and ϕ_{cal} contains sinusoidal hour-of-day and day-of-week features. A learned query token specifies the target, query time, and last positively observed state. A small Transformer produces contextual event features and query representation

$$(\mathbf{h}_1, \dots, \mathbf{h}_K, \mathbf{h}_q) = \text{CTTransformer}(\mathbf{x}_1, \dots, \mathbf{x}_K, \mathbf{x}_q). \quad (9)$$

Unlike a fixed time-bin transition table, this encoder conditions on irregular gaps, repeated negative evidence, and the complete order of observed states. It does not receive oracle mobility type or the activity that generated a hidden transition.

3.5 PERSISTENCE-RELOCATION BELIEF

A direct classifier entangles two physically distinct questions: whether anything changed and where the object went. We instead factorize them. Let s_{last} be the last positive state. The persistence branch predicts

$$\rho_q = P(s_q^o = s_{\text{last}} \mid H_o, t_q) = \sigma(\mathbf{w}_\rho^\top \mathbf{h}_q). \quad (10)$$

For every relocation candidate $i \neq s_{\text{last}}$, a shared pointer head computes

$$a_i = \frac{(W_q \mathbf{h}_q)^\top (W_c \mathbf{g}_i)}{\sqrt{d}} + \psi(\mathbf{h}_q, \mathbf{g}_i), \quad \pi_q(i) = \frac{\exp a_i}{\sum_{j \neq s_{\text{last}}} \exp a_j}, \quad (11)$$

where $\mathbf{g}_i$ is the candidate representation and ψ captures target-candidate compatibility. The prior belief is

$$b_q^-(i) = \begin{cases} \rho_q, & i = s_{\text{last}}, \\ (1 - \rho_q)\pi_q(i), & i \neq s_{\text{last}}. \end{cases} \quad (12)$$

The model is trained end-to-end with a single negative log-likelihood, $\mathcal{L}_{\text{state}} = -\log b_q^-(s_q^o)$. The factorization is therefore an inductive bias, not a separately supervised oracle switch. In static environments it can place mass on persistence; for recurring relocations it can concentrate π_q ; and for irregular objects it can retain a broad distribution.

3.6 VISIBILITY-AWARE BAYESIAN EVIDENCE UPDATE

After navigating to candidate r , the agent obtains detection result z , detector reliability c , and effective visibility $v \in [0, 1]$. A positive instance match assigns high likelihood to r . A negative observation should suppress r only if the target would probably have been detected there. We use

$$P(z = \text{notseen} \mid s_q^o = i, c, v) = \begin{cases} 1 - cv, & i = r, \\ 1, & i \neq r, \end{cases} \quad (13)$$

followed by

$$b_q^+(i) = \frac{b_q^-(i)P(z \mid s_q^o = i, c, v)}{\sum_j b_q^-(j)P(z \mid s_q^o = j, c, v)}. \quad (14)$$

Thus, a clear empty view provides strong negative evidence, whereas an occluded or poorly covered view makes only a small change. The update is parameter free and can be applied repeatedly. Each inspection and its posterior are written back as a provisional memory event, preserving the causal chain used by subsequent decisions.

3.7 BELIEF-GUIDED NAVIGATION AND REPLANNING

Each candidate i has one or more reachable inspection viewpoints $\mathcal{W}_i$. Let $d(x, \mathcal{W}_i)$ be the shortest geodesic distance from the current pose and $\hat{v}_i$ the expected inspection coverage. The high-level policy selects

$$i^* = \arg \max_{i \in \mathcal{C}_o \setminus \mathcal{I}} \frac{b(i)\hat{v}_i}{d(x, \mathcal{W}_i) + \lambda_{\text{inspect}}}, \quad (15)$$

where $\mathcal{I}$ contains sufficiently inspected hypotheses. The fixed low-level navigator executes the route and the evidence update in Equation (14) produces the belief for the next decision. This separation ensures that gains reflect memory and state selection rather than a different locomotion controller.

The same interface instantiates three core tasks. *Predictive Object Navigation* executes only the top-belief candidate and measures first-go success. *Belief-Guided Object Search* follows the probability-cost policy under a candidate budget. *Evidence-Aware Replanning* updates after every qualified view and measures recovery from an incorrect first prediction. For online-dynamic episodes, the model is queried at the candidate arrival time $t + \hat{\tau}_i$ rather than the current time:

$$b_{t+\hat{\tau}_i}^- = F_\theta(H_o, t + \hat{\tau}_i), \quad (16)$$

and replanning repeats after every action chunk. No explicit router chooses between “static” and “dynamic” policies; their behavior is induced by the learned belief and the evidence observed during navigation.

4 EXPERIMENTAL SETUP

4.1 P4D-BENCH

The benchmark summary in Figure 1 reports the full P4D-BENCH release: 54 scenes, 90 scene-days, 239.19k dynamic state changes, and 803.68k task instances. The task pool comprises current-state prediction (16%), object navigation (27%), vision-and-language navigation (29%), and question answering (28%). The controlled experiments below focus on prediction and ObjectNav together with EQA diagnostics; the navigation-focused subset is further decomposed into N1–N4, with N5 used as a cross-scene evaluation subset.

Dynamic household construction. P4D-BENCH is built from continuous household histories, not independently sampled placements. We derive activity priors from longitudinal smart-home records such as CASAS (Cook et al., 2013) and ARAS, and action-object evidence from EPIC-KITCHENS-100 and OPPORTUNITY. These records provide statistics rather than trajectories copied into simulation. Across 54 HSSD scenes (Khanna et al., 2024) and 90 generated scene-days, we freeze resident schedules, room preferences, orderliness, object ownership, and placement habits, then jointly generate activities, resident region trajectories, and object transitions. Frozen distributions make earlier days predictive of later ones, while sampled events and exceptions prevent deterministic scripts. The audited configuration yields 42.6 region transitions per resident-day, 6.56 relocations per personal object-day, 52.6% within-region relocations, and zero terminal collisions. Unsupported parameters are explicitly marked as benchmark-authored.

Object-specific dynamics. Each target follows one of four causal mechanisms (Table 1). Every relocation is attributable to an activity, resident transition, or tidying event. Routine objects follow activity chains; personal objects follow a placed/carried lifecycle with owner-specific drop habits; and irregular destinations depend only on affordance. Audits verify that personal objects do not merely track their owners and that activity or actor identity does not predict irregular destinations beyond affordance.

Physical realization and partial observation. Generated transitions are replayed chronologically in Habitat; only semantically valid, collision-free placements with reachable inspection viewpoints enter the benchmark. A robot patrols using the map, time, travel cost, and its own history, never current object states, resident locations, hidden activities, or generator latents. Region visits produce local snapshots from which observations are derived. For a target and query time, the public input contains only the preceding observations, last-seen state, staleness, calendar context, and legal

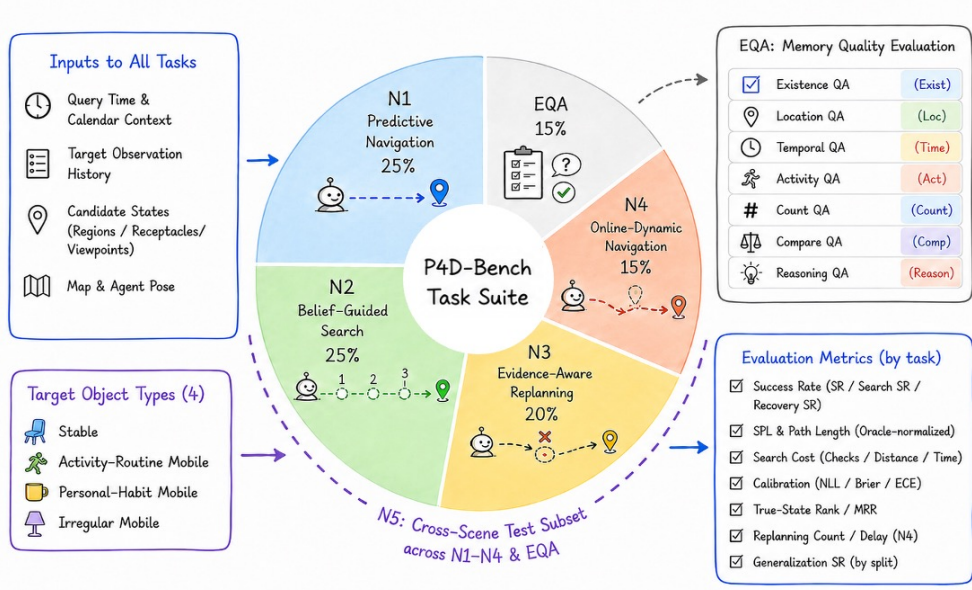


Figure 4: Navigation-focused evaluation protocols in P4D-BENCH. The benchmark contains four navigation protocols (N1–N4) and an embodied question answering (EQA) suite over persistent dynamic world histories. N1 evaluates predictive navigation, N2 belief-guided search, N3 evidence-aware replanning, and N4 online navigation under target dynamics. EQA evaluates memory quality through existence, location, temporal, activity, count, comparison, and compositional reasoning questions. N5 is not a separate task, but a cross-scene evaluation subset applied across N1–N4 and EQA. Target objects are stratified into four mobility types, while all protocols share temporal context, observation history, candidate states, and spatial context. Percentages shown here describe this analysis subset; Figure 1 reports the full task composition.

Table 1: Complementary mobility regimes in P4D-BENCH. Mobility labels are available for stratified evaluation but hidden from the agent.

Regime	Mechanism	Frozen signal	Role
Stable/rare	Infrequent events	Home/return tendency	Persistence control
Routine	Activity state machine	Chain-specific locations	Routine prediction
Personal	Placed/carried lifecycle	Owner placement habit	Long-term memory
Irregular	Affordance sampling	Affordance only	Uncertainty control

candidates. Query times cover both the natural distribution and mechanism-relevant windows, such as stages of a routine or the interval after a personal object is dropped. Current state, hidden transitions, activities, and mobility labels remain private.

From prediction to navigation. The prediction track first evaluates the region/receptacle belief without navigation confounds. Each navigation episode then reuses the same query and adds a start pose, inspection viewpoints, and a budget. N1 tests the Top-1 goal, N2 tests cost-aware search under the full belief, and N3 tests evidence-aware posterior updates and replanning. N4 adds execution-time target motion, while N5 evaluates held-out scenes or households. All methods share candidates, starts, geometry, control, and budgets; only the belief and resulting search order differ.

The full benchmark additionally provides vision-and-language navigation instances over the same evolving histories. They exercise the common memory interface under language-conditioned goals, but are not included in the navigation-focused causal comparisons reported below.

Splits and quality control. We split complete temporal sequences rather than individual queries and reserve disjoint homes for generalization. Audits check future leakage, causal continuity, mobility

Table 2: Prediction and navigation protocols in P4D-BENCH. Navigation episodes are derived from the same target-time queries used for belief evaluation.

Task	Decision signal	Primary measure
P Current-state prediction	Region/receptacle belief	R@ k / NLL
N1 Predictive navigation	Top-1 belief	First-Go SR
N2 Belief-guided search	Full belief + cost	Search SR / cost
N3 Evidence-aware replanning	Updated posterior	Recovery SR
N4 Online-dynamic navigation	Arrival-time belief	Online Recovery SR
N5 Cross-scene generalization	Any protocol above	Held-out SR

behavior, physical validity, and patrol independence. Results are stratified by mobility regime, transition class, staleness, and spatial difficulty; full statistics and audit thresholds are provided in the supplement.

4.2 BASELINES AND ABLATIONS

We compare against four baseline families under identical candidate states, inspection viewpoints, perception inputs, path planner, and navigation budgets.

Non-predictive baselines include Last Seen and Nearest Candidate Search. *Statistical* baselines include Frequency Prior, first-order Markov transitions, time-conditioned transitions, and an instance-hotspot prior. *Learned* baselines include GRU Direct, Transformer Direct, Scene Graph Memory (Kurenkov et al., 2023), a PredictiveGraphs-style filter (Saavedra-Ruiz et al., 2026), and a FlowMaps-style future-location model (Argenziano et al., 2026). Where an external method assumes a different state space, we project its prediction onto the same benchmark candidate set and rank candidates by predicted probability mass, isolating differences in belief estimation rather than downstream planning.

N4 additionally compares against Transit-Aware Planning under its required online target-path input (Dorbala et al., 2026). Oracle Current State provides the navigation upper bound. Oracle Activity supplies the ground-truth current activity and is reported only as a diagnostic rather than as a fair baseline.

For EQA, all methods are evaluated using the same question templates, answer ontology, and answer normalization. The queried information must be recovered from the memory state available to each method. When a learned language readout is required, the readout is held fixed across compared memory methods so that EQA primarily measures memory quality rather than differences in language-model capacity.

The principal ablations remove: continuous-time features; the persistence-relocation factorization; instance-specific history; calendar context; negative evidence; visibility calibration; Bayesian belief updating; or the full belief distribution in favor of Top-1 prediction. *P4D-Belief Prior* predicts once and searches without belief updates, while full P4D-NAV closes the observe-update-replan loop.

4.3 METRICS

We first evaluate the predictive belief independently of downstream navigation using state Top-1/Top- k accuracy, negative log-likelihood (NLL), Brier score, expected calibration error (ECE), true-state rank, mean reciprocal rank (MRR), and persistence AUROC.

For navigation, we report Success Rate (SR), SPL, path length, completion time, First-Go SR, number of inspected candidates, and search cost. N3 additionally reports Recovery SR after an incorrect first prediction and posterior true-state rank improvement after negative evidence. N4 reports Online Recovery SR, replanning count, replanning delay, and excess path length relative to a trajectory-aware oracle. The trajectory-aware oracle is used because a moving target makes the conventional shortest path to a fixed terminal goal ill-defined.

For EQA, we report answer accuracy broken down by existence, location, temporal, activity, count, comparison, and reasoning questions. For questions with structured answer spaces, predictions

Table 3: Main results on routine-world N1–N3 episodes. All methods share the same perception, candidates, inspection viewpoints, and low-level navigator. Lower is better for inspections and distance.

Method	First-Go SR $\uparrow$	Search SR $\uparrow$	SPL $\uparrow$	Inspect. $\downarrow$	Distance $\downarrow$	Recovery SR $\uparrow$
Nearest Candidate	–	–	–	–	–	–
Last Seen	–	–	–	–	–	–
Frequency Prior	–	–	–	–	–	–
Markov Transition	–	–	–	–	–	–
Time-Conditioned Prior	–	–	–	–	–	–
Transformer Direct	–	–	–	–	–	–
PredictiveGraphs-style	–	–	–	–	–	–
FlowMaps-style (Argenziano et al., 2026)	–	–	–	–	–	–
P4D-Belief Prior	–	–	–	–	–	–
P4D-NAV (ours)	–	–	–	–	–	–
Oracle Current State	–	–	–	–	–	–

are normalized to the benchmark ontology before scoring. We additionally report category-macro accuracy so that frequent question types do not dominate the aggregate EQA score.

All aggregate results include bootstrap confidence intervals over episodes and both micro averages and mobility-type macro averages. Results are stratified by world variant, mobility type, transition class (*stay*, *same-region move*, *cross-region move*, *multi-move*, and *return*), staleness interval, spatial difficulty, grounding quality, and generalization split. For EQA, results are additionally stratified by question type.

The key causal contrast is the paired routine gain: the improvement of a method in the routine world relative to its performance under the paired static and random variants. This contrast tests whether gains arise from exploiting persistent temporal regularities rather than from scene geometry, object identity, or generic search behavior.

Implementation and reporting details are provided in Appendices A.7 and A.8.

5 RESULTS AND ANALYSIS

This working draft defines the complete evaluation matrix but does not invent unfinished measurements. Every dash below must be replaced by a reproduced value or a clearly cited published result before submission. The final discussion should be organized around three hypotheses: predictive belief improves the first navigation decision in routine worlds; a calibrated distribution reduces search cost when Top-1 is uncertain; and visibility-aware updates recover efficiently when the prior is wrong.

5.1 DOES PREDICTIVE MEMORY IMPROVE NAVIGATION?

First-Go SR is the cleanest measure of whether a more accurate current-state belief becomes a better physical destination. Search SR and cost then test whether probability mass beyond Top-1 is useful. Recovery SR is evaluated only on episodes in which the initial candidate was incorrect, preventing easy first-attempt successes from hiding poor replanning. The final analysis will report paired episode-level differences and confidence intervals rather than comparing only aggregate means.

5.2 WHEN DOES PREDICTIVE MEMORY HELP?

Stable objects test backward compatibility: a predictive model should not damage a strong last-seen prior. Activity-routine objects test temporal transitions; personal-habit objects test long-term instance memory beyond global activity frequencies; and irregular objects test whether a broad, well-calibrated belief improves search even when Top-1 accuracy has limited headroom. The stratified result matrix is in Table 4; we will additionally plot SR and excess path length against staleness to distinguish genuine temporal reasoning from memorized hotspots.

5.3 DOES THE GAIN COME FROM LEARNABLE DYNAMICS?

The paired-world matrix is reported in Table 5. The expected signature is not uniformly high performance: P4D-NAV should preserve static performance, gain most in routine worlds, and express greater uncertainty in marginal-matched random worlds. An improvement that remains unchanged after routine transitions are randomized would indicate a frequency, category, or route-length shortcut rather than learned dynamics.

5.4 ABLATIONS, GENERALIZATION, AND FAILURE ANALYSIS

The final ablation table will jointly report state NLL and navigation cost, revealing whether a representation improves only classification or also changes useful decisions. In particular, we will compare query-time versus arrival-time belief on N4, prior-only versus Bayesian updating on N3, and structured versus direct heads on cross-scene candidates. Generalization results will separate seen-scene/unseen-household, unseen-scene/unseen-household, unseen object instance, and unseen receptacle instance.

Failures will be categorized as entity association, missing candidate, stale-state prediction, overconfident relocation, insufficient visibility, detector error, unreachable viewpoint, or replanning error. Qualitative trajectories will show the prior distribution, route, inspection coverage, posterior, and next decision at each step. This evidence is necessary to establish that improved navigation follows from the claimed closed-loop mechanism.

6 CONCLUSION

We introduced persistent navigation in evolving worlds, a setting in which an embodied agent must act on a long-term but potentially stale account of its environment. DR-AGENTOS provides a memory-belief-action loop for inferring task-relevant current state, while its P4D-NAV core operationalizes this problem through dynamic object goals: it separates persistence from relocation, represents variable scene-specific candidates, converts visibility-qualified observations into posterior evidence, and uses the evolving distribution to navigate, explore, and replan. P4D-BENCH grounds this question in human-trace-driven HSSD histories and paired static, routine, and random worlds spanning prediction, navigation, and embodied reasoning. The resulting formulation reframes navigation from executing against a remembered snapshot to acting on the world state best supported by accumulated and newly collected evidence. Quantitative conclusions will be added only after the multi-scene navigation evaluation is complete.

7 LIMITATIONS

Although DR-AGENTOS is motivated by general task-relevant world evolution, the current predictive model is operationalized primarily through discrete portable-object states; structural reconfiguration and changing traversability require richer state and action models. P4D-NAV also assumes reliable entity association and an available set of candidate receptacles and inspection viewpoints; pose drift, missed detections, identity switches, or missing candidates can corrupt both prediction and navigation. Its discrete belief represents arbitrary continuous placements less directly than continuous generative models. The source datasets are complementary rather than jointly observed, so P4D-BENCH episodes are human-trace-grounded simulations, not reconstructions of one household. The main protocol freezes the target after the query; N4 is required to test simultaneous target and agent motion. Finally, visibility-aware updates reduce weak negative evidence but cannot guarantee recovery when beliefs and perception fail together or routines shift.

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A SUPPLEMENTARY MATERIAL

A.1 FORMAL EPISODE INTERFACE

Each navigation episode references one causal prediction query. Public fields include schema version, episode and query identifiers, split, task type, scene and household identifiers, world variant, target description, query time, agent start pose, candidate-state identifiers, budgets, and success criteria. Evaluation-private fields include the current state, target pose, valid goal viewpoints, hidden transition class, oracle path, mobility label, and query-time activity. Train and validation releases may expose analysis labels in metadata; the formal test release replaces them with null values and computes grouped metrics in a private evaluator.

The high-level track exposes

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NAVIGATE_TO(state), INSPECT(state), STOP, NOT_FOUND,
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with a fixed navigation backend. The low-level track exposes Habitat-style forward, turn, look, and stop actions. The main paper uses the high-level track to isolate predictive state selection; low-level results measure sensitivity to control and perception.

A.2 EPISODE VALIDITY AND LEAKAGE AUDIT

An episode is retained only if: (i) the target has at least one positive pre-query observation; (ii) the target state is known to the private evaluator and physically instantiated without collision; (iii) at least one valid viewpoint lies on the NavMesh; (iv) the start is on the NavMesh, at least 3 meters from the target, and does not reveal it; and (v) all public histories terminate at or before the query. A successful stop must be within 1 meter geodesic distance of a valid viewpoint, observe the target above a frozen visible-fraction threshold, and identify the correct instance or category.

We audit serialized examples for future timestamps, hidden event fields, filenames or indices correlated with answers, duplicate event groups across splits, and inconsistent paired variants. Static, routine, and random members of a pair must have identical public query fields. Invalid simulator episodes are reported separately and are not counted as ordinary failures.

A.3 MOBILITY-TYPE ASSIGNMENT

The frozen rule is

$$\text{type}(o) = \begin{cases} \text{stable,} & r_{\text{move}}(o) < \tau_{\text{move}}, \\ \text{activity,} & G_{\text{act}}(o) \geq \tau_{\text{act}}, \\ \text{personal,} & C_{\text{habit}}(o) \geq \tau_{\text{habit}}, \\ \text{irregular,} & \text{otherwise.} \end{cases} \quad (17)$$

All thresholds are selected on training data before navigation evaluation. When the generator specifies a mobility profile, the same statistics are used to verify that realized trajectories exhibit the intended behavior. Every group must cover several categories, instances, and scenes, and no category may uniquely reveal one mobility type.

A.4 METRIC DEFINITIONS

Let S_i be episode success, L_i the executed path, and L_i^* the oracle shortest path to the query-time goal under the same start. Standard SPL is

$$\text{SPL} = \frac{1}{N} \sum_{i=1}^N S_i \frac{L_i^*}{\max(L_i, L_i^*)}. \quad (18)$$

First-Go SR counts success when only the initial predicted candidate may be inspected. Recovery SR conditions on an incorrect first candidate and measures whether the target is subsequently found. Excess path is $L_i - L_i^*$ for static-after-query episodes. For online-dynamic episodes, the oracle is trajectory aware and may intercept the target; we therefore report both success within the common budget and $L_i - L_{i,\text{dyn}}^*$.

Calibration metrics operate on the full candidate belief before navigation and after each update. ECE bins maximum confidence, Brier score evaluates all candidates, and true-state rank avoids hiding useful changes below Top-1. Search-cost curves plot success against allowed inspections and travelled distance rather than reporting one arbitrary budget.

A.5 FULL BASELINE SPECIFICATION

Last Seen assigns all mass to the latest positive state. Frequency Prior estimates $P(s \mid o)$ from training data. Markov Transition estimates $P(s_{k+1} \mid s_k, o)$; Time-Conditioned Prior additionally conditions on public hour and weekday bins. Instance Hotspot uses only the target instance’s training trajectory. GRU Direct and Transformer Direct receive the same event tokens and candidate encoder as P4D-NAV but directly normalize candidate logits without the persistence factorization. P4D-Belief Prior uses Equation (12) once at episode start. Full P4D-NAV applies Equation (14) and Equation (15) after each inspection.

The Oracle Activity diagnostic may use the hidden activity and source location but is excluded from fair rankings. Oracle Current State receives the private target state and measures remaining perception and navigation error. External predictive methods are adapted only through their public inputs; any use of privileged simulator state must be identified and excluded from the main comparison.

A.6 DISAGGREGATED RESULT TEMPLATES

A.7 IMPLEMENTATION DETAILS

Unless otherwise noted, the history encoder has three Transformer layers, hidden width 128, four attention heads, and at most 64 events. The predictive model contains approximately –million trainable parameters and is optimized with AdamW using validation NLL for checkpoint selection. Candidate encoders are shared across scenes. The main high-level track fixes candidate viewpoints

Table 4: Search SR by hidden mobility type. Types are determined from training history and withheld from the agent. Macro average weights the four groups equally.

Method	Stable	Activity routine	Personal habit	Irregular	Macro
Last Seen	–	–	–	–	–
Time-Conditioned	–	–	–	–	–
Instance Hotspot	–	–	–	–	–
Transformer Direct	–	–	–	–	–
P4D-NAV	–	–	–	–	–

Table 5: Paired-world navigation results. Variants share all observed setup variables and differ only in hidden transitions.

Method	Static SR	Routine SR	Random SR	Routine gain over Last Seen
Last Seen	–	–	–	–
Markov Transition	–	–	–	–
Transformer Direct	–	–	–	–
P4D-NAV	–	–	–	–

and low-level planning; a Habitat low-level action track evaluates the complete embodied stack. Final learning rate, batch size, epoch count, hardware, simulator version, detector, visibility thresholds, and repeated-run protocol will be frozen before test evaluation.

A.8 REPRODUCIBILITY CHECKLIST

The completed release will report:

- source dataset versions, licenses, mappings, filters, and canonical-record statistics;
- HSSD and Habitat versions, scene identifiers, object assets, placement checks, and NavMesh settings;
- split manifests, paired-example identifiers, checksums, and leakage-audit output;
- model width, layers, attention heads, time features, history length, optimizer, learning rate, epochs, seeds, and checkpoint rule;
- detector and depth inputs, visibility computation, candidate and action budgets, evidence likelihoods, and planner hyperparameters;
- hardware, wall-clock latency, memory growth, repeated-run aggregation, and confidence intervals; and
- reproduced versus quoted baseline values, grouped results, failure labels, and all component-aligned ablations.

A.9 SUBMISSION COMPLIANCE NOTES

This source is configured for anonymous ICLR 2027 submission: \iclrfinalcopy remains disabled, author identities and affiliations are absent, and references precede the appendix. Before submission, authors should remove all result placeholders, keep the main text within the conference page limit, and scan figure metadata, linked repositories, PDF properties, acknowledgements, and supplementary files for identifying information.

A.10 ETHICS, REPRODUCIBILITY, AND AI USE

Ethics. Persistent visual memory may capture people or private spaces, while an incorrect belief may induce unnecessary motion or unsafe behavior. Deployments should obtain consent, enforce retention and access controls, preserve provenance, and retain platform-specific collision avoidance and emergency stops outside the learned model. Our benchmark compiles existing records into simulated HSSD episodes and introduces no new human-subject study.

Reproducibility. Section 3 specifies the task, memory, predictor, evidence update, and navigation policy. Section 4 defines episode construction, leakage controls, baselines, and metrics. An anonymized code and data snapshot will accompany the review submission.

AI use. Generative AI tools assisted manuscript editing, LaTeX checks, and local engineering. They did not generate benchmark observations, ground-truth labels, or reported scores. The authors reviewed all assisted material and remain responsible for the content and artifacts.